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CLASSICAL

BECAUSE $|x_+, x_+\rangle = |x_+\rangle \otimes |x_+\rangle$
 $|x_+\rangle = \frac{1}{\sqrt{2}} [|z_+\rangle + |z_-\rangle]$

$$|x_+, x_+\rangle = \frac{1}{2} [|z_+, z_+\rangle + |z_-, z_+\rangle + |z_+, z_-\rangle + |z_-, z_-\rangle]$$

PROB. BOTH HEADS X

$$P = \frac{1}{2} \quad |z_+, z_+\rangle$$

$$P = \frac{1}{2} \quad |z_-, z_-\rangle$$

$$P(|x_+, x_+\rangle) = |\langle z_+, z_+ | x_+, x_+ \rangle|^2 = \left| \frac{1}{2} + 0 + 0 + 0 \right|^2 = \frac{1}{4}$$

$$\underbrace{\frac{1}{2} \cdot \frac{1}{2}}_{P=1/2} + \underbrace{\frac{1}{2} \cdot \frac{1}{2}}_{P=1/2} = \frac{1}{4}$$

4 POSSIBILITIES

$|x_+, x_+\rangle$

$|x_-, x_+\rangle$

$|x_+, x_-\rangle$

$|x_-, x_-\rangle$

EQUALLY PROBABLE AND
TOTAL PROB. IS 1 $\Rightarrow$ $\frac{1}{4}$ PROB
WITH ALL OF THEM

QUANTUM

$$| \eta \rangle = \frac{1}{\sqrt{2}} [|z_+, z_+\rangle + |z_-, z_-\rangle]$$

$$|x_+, x_+\rangle = \frac{1}{2} [|z_+, z_+\rangle + |z_-, z_+\rangle + |z_+, z_-\rangle + |z_-, z_-\rangle]$$

$$P(|x_+, x_+\rangle) = |\langle \eta | x_+, x_+ \rangle|^2 = \left| \frac{1}{\sqrt{2}} \cdot \frac{1}{2} (1+1) \right|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

DIFFERENT PROB

$$P(|x_-, x_-\rangle) = \dots = \frac{1}{2}$$

NO WAY A. FIND SPIN UP
AND B. FIND SPIN DOWN AND VICE
VERSA

$$\text{THEN } P(|x_-, x_+\rangle) = 0 = P(|x_+, x_-\rangle)$$

$\frac{1}{2} \quad y^+$

$$\frac{1}{\sqrt{2}} (|z_+, z_+\rangle$$

$\frac{1}{2} \quad \frac{1}{2}$

X IS NOT A SPECIAL DIRECTION, SAME RESULT ALSO ON Y OR ANY
DIRECTION IN THE XY

OUR RESULTS DEPENDS ON THE INTERFERENCE

$$\frac{1}{\sqrt{2}} (|z_+, z_+\rangle - |z_-, z_-\rangle)$$

$$\frac{1}{\sqrt{2}} [|z_+, z_+\rangle + |z_-, z_-\rangle]$$

$$P(|z_+, z_+\rangle) = \frac{1}{2}$$

$$P(|z_-, z_-\rangle) = \frac{1}{2}$$

$$\frac{1}{\sqrt{2}} [|z_+, z_+\rangle - |z_-, z_-\rangle]$$

$$P(|z_+, z_+\rangle) = \frac{1}{2}$$

$$P(|z_-, z_-\rangle) = \frac{1}{2}$$

$$P(|x_+, x_+\rangle) = 1/2$$

$$P(|x_+, x_+\rangle) =$$

$$P(|z_-, z_+\rangle) = |\langle \eta | z_-, z_+\rangle|^2$$

$$|\langle \eta | x_+, x_+\rangle|^2 = \left| \frac{1}{\sqrt{2}} (\langle z_+ z_+ | - \langle z_- z_- |) \right| \frac{1}{\sqrt{2}} \left[|z_+, z_+\rangle + |z_-, z_+\rangle + |z_-, z_+\rangle + |z_+, z_+\rangle \right] \right|^2$$

$$= \frac{1}{8} |1 - 1|^2 = 0$$

$$\frac{1}{\sqrt{2}} (|1, 1\rangle + |0, 0\rangle)$$

$$x \ x \ x \ y \ y \ x \ x \ y \ x \ x$$

WE DEFINE THE **CONDITIONAL PROBABILITY** $P(b|\omega) = \frac{P(\omega, b)}{P(\omega)}$

$$1) \forall \omega \sum_b P(b|\omega) = 1$$

$$2) \omega \text{ AND } b \text{ ARE INDEPENDENT RANDOM VARIABLES} \Leftrightarrow \forall \omega, b \ P(b|\omega) = P(b)$$

$$P(b|\omega) = \frac{P(\omega, b)}{P(\omega)}$$

REMEMBER

$$P(\omega) = \sum_b P(\omega, b)$$

$$P(b) = \sum_{\omega} P(\omega, b)$$

$$1) \forall \omega \sum_b P(b|\omega) = 1$$

$$\hookrightarrow \sum_b \frac{P(\omega, b)}{P(\omega)} = \frac{1}{P(\omega)} \sum_b P(\omega, b) = \frac{1}{P(\omega)} \cdot P(\omega) = 1$$

$$2) \omega \wedge b \text{ ARE INDEPENDENT} \Leftrightarrow \forall \omega, b \ P(b|\omega) = P(b)$$

$$\bullet \Rightarrow P(\omega, b) \stackrel{\text{DEF. OF INDEPENDENT}}{=} P(\omega) P(b)$$

$$P(\omega|b) = \frac{P(\omega, b)}{P(b)} = \frac{P(\omega) P(b)}{P(b)} = P(\omega)$$

• $\Leftarrow$

$$P(b|a) = P(b)$$

$$P(a,b) = P(b|a) \cdot P(a) = \underbrace{P(b) P(a)}_{\text{INDEPENDENT}}$$

CONDITIONAL STATE

$$|\psi^{(A,B)}\rangle = \sum_{\omega} |\omega\rangle \otimes |\phi_{\omega}\rangle \quad \text{WHERE } |\phi_{\omega}\rangle \text{ ARE NON-NORMALIZED VECTORS IN } \mathcal{H}^{(B)}$$

$$P(b|a) = \frac{P(a,b)}{P(a)} = \frac{|\langle b|\phi_a\rangle|^2}{P(a)} = \frac{|\langle b|\phi_a\rangle|^2}{|\langle \phi_a|\phi_a\rangle|^2}$$

CONDITIONAL STATE

$$|\phi'_{\omega}\rangle = \frac{1}{\sqrt{P(a)}} |\phi_{\omega}\rangle$$

$$P(b|a) = |\langle b|\phi'_{\omega}\rangle|^2$$

SUPPOSE THE STATE OF A SYSTEM Q ARISES BY A RANDOM PROCESS, SO THAT Q IS IN THE STATE $|\psi_{\alpha}\rangle$ WITH PROBABILITY P_{α}

$$\{|\psi_{\alpha}\rangle, P_{\alpha}\}$$

• THE STATES $|\psi_{\alpha}\rangle$ ARE STATE (NORMALIZED) VECTORS

• NOTE THAT THE STATE $|\psi_{\alpha}\rangle$ DO NOT NEED TO BE ORTHOGONAL

$$\sum_{\alpha} P_{\alpha} = 1$$

$$\{ |\psi_a\rangle, p_a \} \quad \langle A \rangle = ?$$

FOR EVERY STATE $|\psi_a\rangle$ WE KNOW THAT $\langle A \rangle_a = \langle \psi_a | A | \psi_a \rangle$

$$\langle A \rangle_a = \langle \psi_a | A | \psi_a \rangle = \text{Tr} \{ |\psi_a\rangle \langle \psi_a | A \}$$

DENSITY OPERATOR

$$\rho = \sum_a p_a |\psi_a\rangle \langle \psi_a|$$

$$\langle A \rangle = \text{Tr}(\rho A)$$

THE DENSITY OPERATOR ALLOWS ONE TO CALCULATE ANY OF THE EXPECTATION VALUES, THUS IT DESCRIBES THE MIXED STATE

MIXED AND PURE STATE

$$\rho = \sum_a p_a |\psi_a\rangle \langle \psi_a|$$

MIXED STATE

$$\rho = |\psi\rangle \langle \psi|$$

PURE STATE

SOME EXAMPLE

$\mathcal{B} = \{ |0\rangle, |1\rangle \}$ 1 QUBIT

EX: PURE STATE $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$

CHECK IF NORMALIZED

$$\langle \psi | \psi \rangle = \left| \frac{1}{2} \right|^2 + \left| \frac{\sqrt{3}}{2} \right|^2 = \frac{1}{4} + \frac{3}{4} = 1 \quad \checkmark$$

DENSITY OPERATOR

$$\rho[|\psi\rangle\langle\psi|] = \left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \right) \left(\frac{1}{2}\langle 0| + \frac{\sqrt{3}}{2}\langle 1| \right) = \frac{1}{4}|0\rangle\langle 0| + \frac{\sqrt{3}}{4}|0\rangle\langle 1| + \frac{\sqrt{3}}{4}|1\rangle\langle 0| + \frac{3}{4}|1\rangle\langle 1|$$

ONE A MATRIX REPRESENTATION IT IS 2x2 BECAUSE HILBERT SPACE OF SIZE 2

$$\rho = \begin{bmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{bmatrix}$$

$\rho_{ij} = \langle i | \rho | j \rangle$

MUST BE A BASIS

$$\rho_{11} = \langle 0 | \rho | 0 \rangle = 1/4$$

$$\rho_{22} = \langle 1 | \rho | 1 \rangle = 3/4$$

$$\rho_{21} = \langle 1 | \rho | 0 \rangle = \sqrt{3}/4$$

$$\rho_{12} = \langle 0 | \rho | 1 \rangle = \sqrt{3}/4$$

MIXED STATE

$$P = 1/3 \quad |0\rangle \rightarrow |0x0\rangle$$

$$P = 2/3 \quad |1\rangle \rightarrow |1x1\rangle$$

$$P = 1/3 |0x0\rangle + 2/3 |1x1\rangle$$

$$P = \sum_a P_a |\psi_a\rangle \langle \psi_a| = \frac{1}{3} |0x0\rangle \langle 0x0| + \frac{2}{3} |1x1\rangle \langle 1x1|$$

REPRESENTATION BASIS $B = \{|0\rangle, |1\rangle\}$ $P = \begin{bmatrix} 1/3 & 0 \\ 0 & 2/3 \end{bmatrix}$

ex: 8.2

$B = \{|0\rangle, |1\rangle\}$ $| \psi \rangle = \alpha |0\rangle + \beta |1\rangle$ $|\alpha|^2 + |\beta|^2 = 1 \Rightarrow \langle \psi | \psi \rangle = 1$

WRITE A MIXED STATE

$$P = |\alpha|^2 |0x0\rangle \langle 0x0| + |\beta|^2 |1x1\rangle \langle 1x1|$$

$$P = \sum_a P_a |\psi_a\rangle \langle \psi_a|$$

TEST

$$P_{\psi}(0) = |\langle 0 | \psi \rangle|^2 = |\alpha|^2$$

$$P_{\psi}(0) = \langle 0 | P | 0 \rangle = \text{Tr}(P |0x0\rangle \langle 0x0|)$$

$$\downarrow$$

$$\langle 0 | |\alpha|^2 |0x0\rangle \langle 0x0| + |\beta|^2 |1x1\rangle \langle 1x1| | 0 \rangle = |\alpha|^2$$

$$P | \psi \rangle = |\psi\rangle \langle \psi| = (\alpha |0\rangle + \beta |1\rangle)(\alpha^* \langle 0| + \beta^* \langle 1|) = |\alpha|^2 |0x0\rangle \langle 0x0| + \alpha \beta^* |1x0\rangle \langle 0x0| + \beta \alpha^* |0x1\rangle \langle 1x1| + |\beta|^2 |1x1\rangle \langle 1x1|$$

$B = \{|0\rangle, |1\rangle\}$ $P_{|\psi\rangle} = \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \\ \beta \alpha^* & |\beta|^2 \end{bmatrix}$

$$P_{mix} = |\alpha|^2 |0x0\rangle \langle 0x0| + |\beta|^2 |1x1\rangle \langle 1x1|$$

$$P_{mix} = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix}$$

$$|x\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$\langle x \rangle = \langle x_+ | P | x_+ \rangle (+1) + \langle x_- | P | x_- \rangle (-1)$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Tr}(P_{|\psi\rangle} X) = \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \\ \beta \alpha^* & |\beta|^2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \alpha \beta^* + \beta \alpha^* = 2 \text{Re} \{ \alpha \beta^* \}$$

$$\text{Tr}(P_{mix} X) = \text{Tr} \left\{ \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\} = 0$$

$$\langle x \rangle = \langle 0 | x | 0 \rangle = 0$$

$$\langle 1 | x | 1 \rangle = 0$$

↑ obvious

$$\rho = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$$

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$\langle H \rangle \neq 0 \wedge \neq 1$$

$$\langle - \rangle \neq 0 \wedge \neq 1$$

$$\rho = \frac{1}{2} I$$